# Factor Analysis of Work Engagement

## Women & Workforce Engagement

EFA & CFA of the 9-Item Utrecht Work Engagement Scale (UWES-9)
in a Multi-Occupational Female Sample — With Full Mathematical Treatment

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Primary Study:

**Willmer, M., Westerberg Jacobson, J. & Lindberg, M. (2019)**

Frontiers in Psychology, 10, Article 2771

DOI: 10.3389/fpsyg.2019.02771

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# Presentation Agenda

Structure reflects greater emphasis on EFA over CFA per professor feedback

**FOCUS AREAS:**

- EFA (Mathematical Deep-Dive)
- Introduction / Background
- CFA & Conclusion (Summary Level)

1. **Introduction & Background**
2. **The UWES-9 Instrument & Data**
3. **Mathematical Foundations of EFA**
4. **Correlation Matrix & Eigenvalues**
5. **KMO & Bartlett's Test — Math**
6. **Factor Extraction & Loadings**
7. **Factor Retention Criteria**
8. **Promax Rotation — EFA Results**
9. **CFA: Model Fit Summary**
10. **Discussion, Conclusion & References**

# Introduction & Background

Work Engagement in Women | UWES-9 | Research Gap

## What Is Work Engagement?

**Vigor**
High energy, resilience, willingness to invest effort

**Dedication**
Significance, enthusiasm, inspiration, pride

**Absorption**
Fully engrossed in work; time passes quickly

## UWES-9 Scale

- 9 items rated 0–6 (Never → Always)
- 3 items per subscale (Vigor / Dedication / Absorption)
- Split-half design: EFA n = 341 | CFA n = 342 (Total N = 702)
- All-female, multi-occupational, Swedish sample, age 26–37

## Research Gap & Purpose

**PROBLEM**

Kulikowski (2017) reviewed 21 studies: 3 confirmed 1-factor, 3 confirmed 3-factor, 4 found both equivalent — no consensus.

**GAP**

No all-female, multi-occupational Swedish sample existed. Gender may change how engagement is structured.

**SOLUTION**

Apply EFA then CFA on a split sample of n = 702 Swedish women to resolve the 1-factor vs. 3-factor debate.

## Key Prior Findings

- Hallberg & Schaufeli (2006): n=186, 1-factor & 3-factor equal fit
- Nerstad et al. (2010): n=1,266, 3-factor but r(F)=0.79–0.84
- Seppala et al. (2009): n=9,404, both structures reasonable

# The UWES-9 Instrument & Sample

Items, Subscales, and Descriptive Statistics

## UWES-9 Items, Subscale Assignment & Descriptive Statistics

| Item | Subscale | Wording (abbreviated) | Mean | SD | Skew | Kurt |
|---|---|---|---|---|---|---|
| 1 | Vigor | At work, I feel full of energy | 3.97 | 1.28 | -0.36 | -0.29 |
| 2 | Vigor | At work, I feel strong and vigorous | 3.88 | 1.30 | -0.32 | -0.35 |
| 3 | Dedication | My work is full of meaning and purpose | 4.36 | 1.28 | -0.62 | 0.10 |
| 4 | Dedication | I am enthusiastic about my job | 4.22 | 1.35 | -0.57 | -0.21 |
| 5 | Vigor | When I wake up, I feel like going to work | 4.02 | 1.45 | -0.41 | -0.52 |
| 6 | Absorption | I feel happy when I am working intensely | 4.09 | 1.28 | -0.49 | -0.08 |
| 7 | Dedication | I am proud of the work that I do | 4.62 | 1.23 | -0.74 | 0.18 |
| 8 | Absorption | I am immersed in my work | 3.98 | 1.26 | -0.30 | -0.34 |
| 9 | Absorption | I get carried away when I am working | 3.84 | 1.33 | -0.14 | -0.64 |

**Vigor** **Dedication** **Absorption**

Overall Mean = 4.06 / 6.0 | Cronbach's α = 0.947 | Skewness & Kurtosis within ±2.0 → ML estimation appropriate

# Mathematical Foundations of Factor Analysis

The Core Model: $X = \Lambda F + \varepsilon$

**The Factor Analysis Model**

$$X = \Lambda F + \varepsilon$$

**X**

**Observed Data Matrix**
**($p \times n$: items × subjects)**

- $p$ = 9 UWES-9 items
- $n$ = 341 (EFA sample)
- $X_i$ = standardized score of item i

**Λ**

**Factor Loading Matrix**
**($p \times m$: items × factors)**

- $\lambda_{ij}$ = loading of item i on factor j
- $m$ = number of factors (1 or 3)
- Estimated by ML extraction

**F**

**Common Factor Matrix**
**($m \times n$: factors × subjects)**

- Latent (unobserved) factors
- $F \sim MVN(0, I)$ assumed
- Factors are orthogonal initially

**ε**

**Unique Factor Matrix**
**($p \times n$: unique variances)**

- Item-specific error variance
- $\varepsilon \sim N(0, \Psi)$ where $\Psi$ diagonal
- $u^2 = 1 - h^2$ (uniqueness)

**Implied Covariance Structure: $\Sigma = \Lambda\Lambda T + \Psi \implies R = \Lambda\Lambda T + \Psi$ (standardized form)**

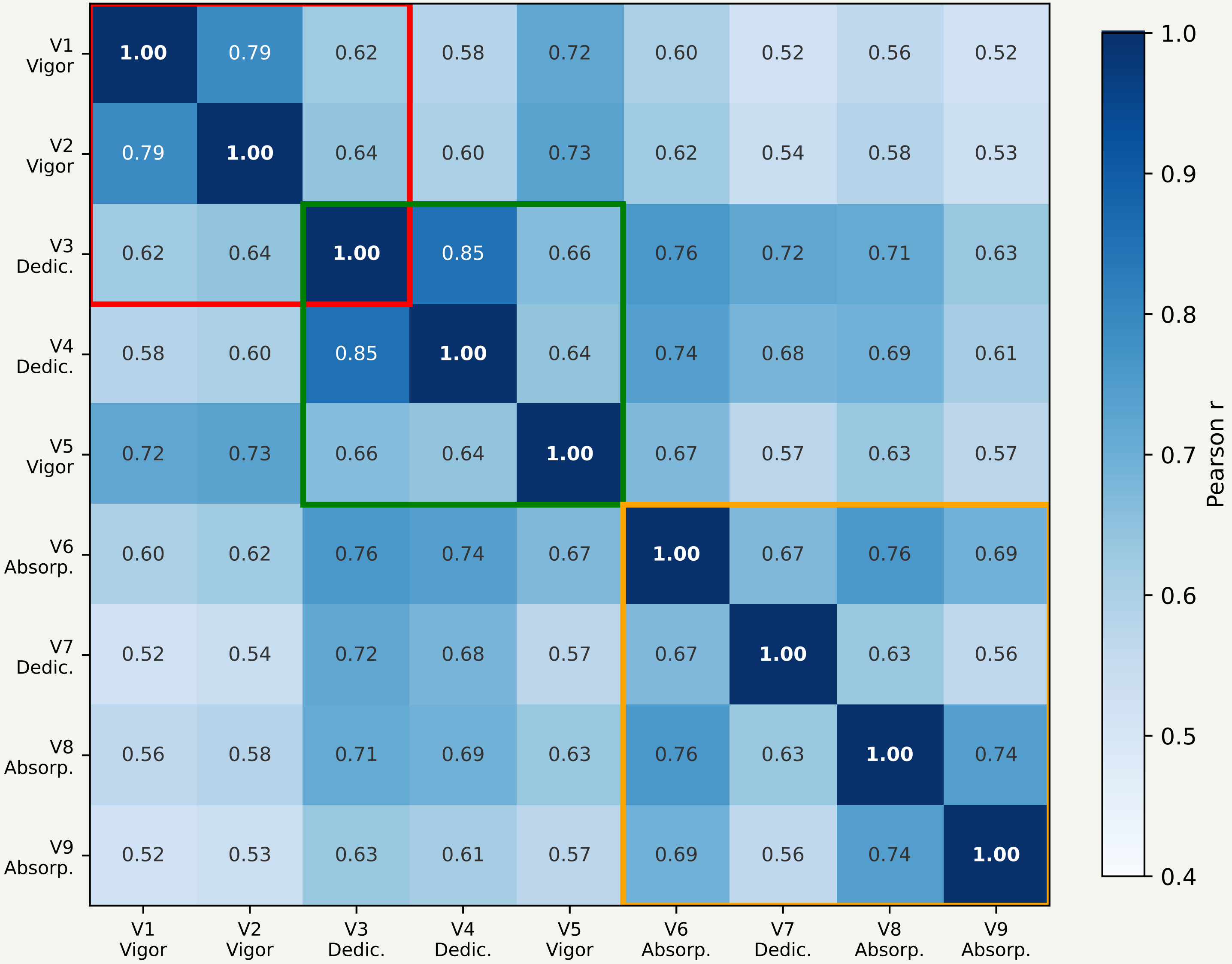
$\Sigma$ = population covariance matrix | $\Lambda\Lambda T$ = common variance (communality) | $\Psi$ = unique variance diagonal matrix

Communality of item i: $h^2_i = \lambda^2_{i1} + \lambda^2_{i2} + \ldots + \lambda^2_{im}$ (sum of squared loadings across all m factors)

# Step 1: Inter-Item Correlation Matrix R

EFA begins by computing R — the standardized covariance matrix of all 9 UWES items (EFA n = 341)

**Correlation Matrix R (p×p, p=9)**
**Coloured borders = theoretical subscale groupings**




| | V1 Vigor | V2 Vigor | V3 Dedic. | V4 Dedic. | V5 Vigor | V6 Absorp. | V7 Dedic. | V8 Absorp. | V9 Absorp. |
|---|---|---|---|---|---|---|---|---|---|
| V1 Vigor | 1.00 | 0.79 | 0.62 | 0.58 | 0.72 | 0.60 | 0.52 | 0.56 | 0.52 |
| V2 Vigor | 0.79 | 1.00 | 0.64 | 0.60 | 0.73 | 0.62 | 0.54 | 0.58 | 0.53 |
| V3 Dedic. | 0.62 | 0.64 | 1.00 | 0.85 | 0.66 | 0.76 | 0.72 | 0.71 | 0.63 |
| V4 Dedic. | 0.58 | 0.60 | 0.85 | 1.00 | 0.64 | 0.74 | 0.68 | 0.69 | 0.61 |
| V5 Vigor | 0.72 | 0.73 | 0.66 | 0.64 | 1.00 | 0.67 | 0.57 | 0.63 | 0.57 |
| V6 Absorp. | 0.60 | 0.62 | 0.76 | 0.74 | 0.67 | 1.00 | 0.67 | 0.76 | 0.69 |
| V7 Dedic. | 0.52 | 0.54 | 0.72 | 0.68 | 0.57 | 0.67 | 1.00 | 0.63 | 0.56 |
| V8 Absorp. | 0.56 | 0.58 | 0.71 | 0.69 | 0.63 | 0.76 | 0.63 | 1.00 | 0.74 |
| V9 Absorp. | 0.52 | 0.53 | 0.63 | 0.61 | 0.57 | 0.69 | 0.56 | 0.74 | 1.00 |

## What R Tells Us

**General Formula**

$$r_{ij} = \frac{\sum_{k=1}^{n}(x_{ik}-\bar{x}_i)(x_{jk}-\bar{x}_j)}{\sqrt{\sum(x_{ik}-\bar{x}_i)^2 \cdot \sum(x_{jk}-\bar{x}_j)^2}}$$

**Inter-item range:** 0.52 – 0.85 (all highly positive)

**Intra-subscale avg:** ~ 0.72–0.84 (e.g., V1–V2 = 0.79)

**Cross-subscale avg:** ~ 0.56–0.68 (lower but still high)

**Bartlett's $\chi^2$:** $p < 0.001$ → R ≠ I (not identity)

**KMO:** 0.922 (Marvellous — Kaiser 1974)

**Key Insight**

High inter-item correlations (0.52–0.85) with no clear block structure suggest that ALL 9 items share a single dominant common factor.

**Eigenvalue Hint**

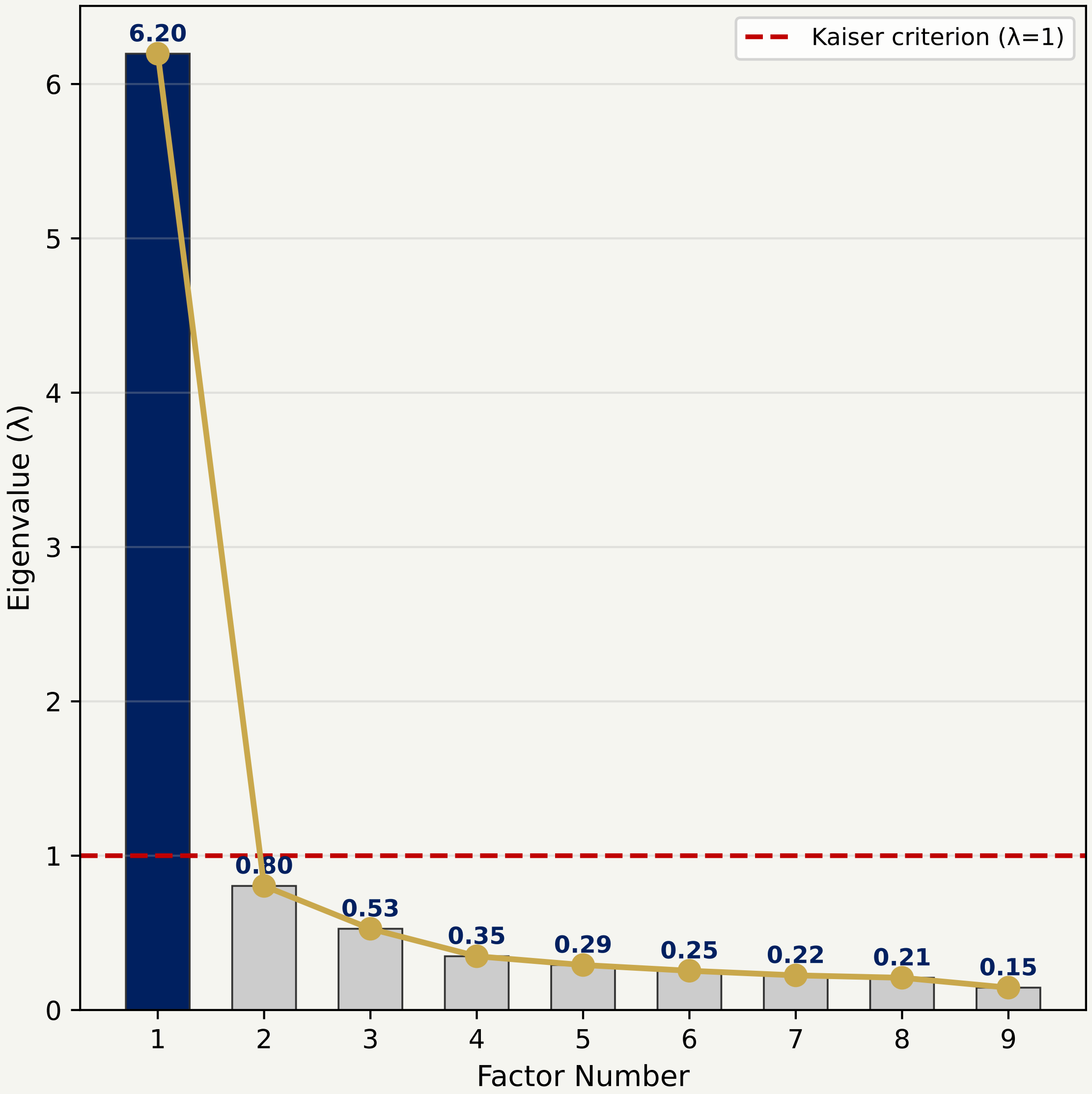
**$\lambda_1 = 6.196$ $\lambda_2 = 0.804$ $\lambda_3 = 0.526$ $\lambda_4 = 0.348$ …**

$\lambda_1 / \Sigma\lambda = 0.688$ = 68.8% of trace
(≈ equivalent to observed > 70% variance explained by Factor 1)

# Step 2: Eigenvalue Decomposition of R

$R = V \Lambda V^T$ — Spectral Decomposition reveals latent factor structure

**Scree Plot (Approximate, UWES-9 EFA n=341)**




## Spectral Decomposition Mathematics

$\mathbf{R} = \mathbf{V\Lambda V}^T$ where $V$ = eigenvectors, $\Lambda = \mathrm{diag}(\lambda_i)$

| Factor | λ | % Var. | Cum. % | >1? |
|---|---|---|---|---|
| 1 | 6.196 | 68.8% | 68.8% | ✓ |
| 2 | 0.804 | 8.9% | 77.8% | × |
| 3 | 0.526 | 5.8% | 83.6% | × |
| 4 | 0.348 | 3.9% | 87.5% | × |
| 5 | 0.292 | 3.2% | 90.7% | × |
| 6 | 0.254 | 2.8% | 93.6% | × |

**Percent Variance Explained:**

**Factor 1: $\lambda_1/\mathrm{trace}(R) = 6.196/9 = 68.8\%$**

The paper reports Factor 1 explains > 70% of variance, consistent with a dominant single-factor structure.

# Step 3: Sampling Adequacy — KMO & Bartlett's Test

Before extracting factors, verify the data is suitable for factor analysis

## Kaiser-Meyer-Olkin (KMO) Measure

$$\mathrm{KMO} = \frac{\sum_{i \neq j} r_{ij}^2}{\sum_{i \neq j} r_{ij}^2 + \sum_{i \neq j} q_{ij}^2}$$

where $r_{ij}$ = observed correlation, $q_{ij}$ = partial correlation (controlling all other variables)

**KMO = 0.922**

*"Marvellous" (Kaiser, 1974)*

**KMO Interpretation Scale (Kaiser, 1974)**

| | | |
|---|---|---|
| 0.90 – 1.00 | **Marvellous** | **← 0.922** |
| 0.80 – 0.89 | Meritorious | |
| 0.70 – 0.79 | Middling | |
| 0.60 – 0.69 | Mediocre | |
| < 0.60 | Unacceptable | |

## Bartlett's Test of Sphericity

$H_0$: R = I (no correlations exist)

$H_1$: R ≠ I (correlations exist)

$$\chi^2 = -\left[(n-1) - \frac{2p+5}{6}\right]\ln|\mathbf{R}|$$

df = p(p − 1) / 2 = 9(8) / 2 = 36

**$\chi^2$ (Bartlett) = significant**

**$p < 0.001$ df = 36**

Substituting n = 341, p = 9, |R| computed from the correlation matrix above:

n − 1 = 340

(2p+5)/6 = (18+5)/6 = 3.83

Adjusted n = 340 − 3.83 = 336.17

$\chi^2 = -336.17 \times \ln|R| \rightarrow p < 0.001$

**Both Tests: PASS**

Data is suitable for factor analysis.

KMO = 0.922 → "Marvellous"

Bartlett $p < 0.001$ → Reject $H_0$

# Step 4: Maximum Likelihood Factor Extraction

ML minimizes the discrepancy between observed R and model-implied $\Sigma(\Lambda, \Psi)$

## ML Fit Function (Discrepancy Function)

$$F_{ML} = \ln|\hat{\Sigma}| - \ln|\mathbf{R}| + \mathrm{tr}(\mathbf{R}\hat{\Sigma}^{-1}) - p \quad \text{where} \quad \hat{\Sigma} = \Lambda\Lambda^\top + \Psi$$

Minimized iteratively over $\Lambda$ and $\Psi$; at convergence $\hat{\Sigma} \approx R \rightarrow$ chi-sq = $(n-1) \times F_{ml}$

**ML Model Fit Test**
**(1-Factor Solution)**

**$\chi^2 = 332.43$**

**df = 27, $p < 0.001$**

**Chi-sq formula:**

$$\chi^2 = (n-1) \times F_{ML}$$

df = p(p+1)/2 − (pm − m(m−1)/2 + m)

= 45 − (9 − 1 + 1 + 1) = 27

$= (341 - 1) \times F_{ml}$

$= 340 \times 0.978$

**= 332.43**

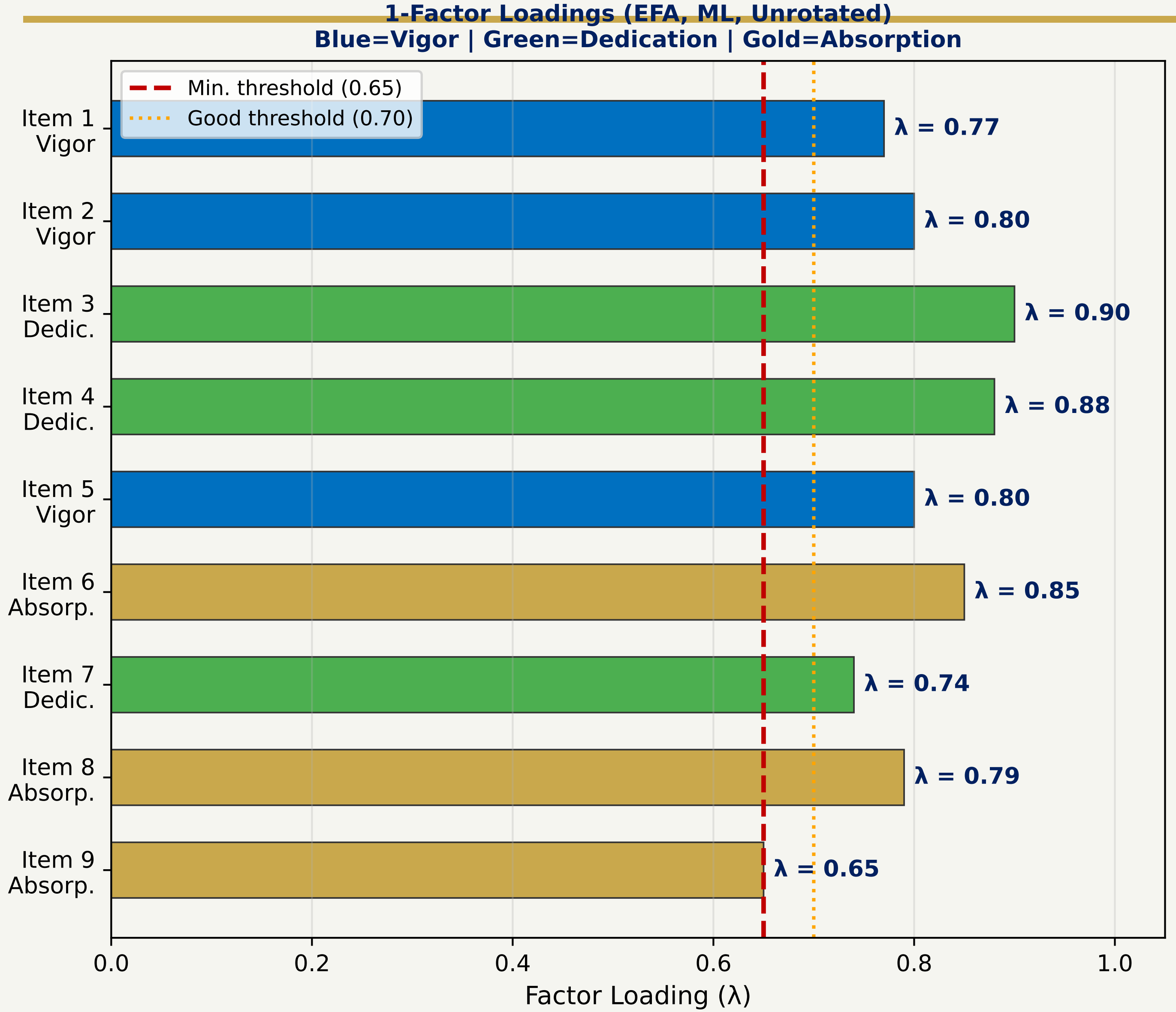
### Communalities ($h^2$) — Variance Explained by the Common Factor

| Item | Loading ($\lambda$) | $h^2 = \lambda^2$ | Uniqueness $u^2 = 1-h^2$ | Variance Bar |
|---|---|---|---|---|
| Item 1 Vigor | 0.77 | 0.59 | 0.41 | $h^2$=0.59 $u^2$=0.41 |
| Item 2 Vigor | 0.80 | 0.64 | 0.36 | $h^2$=0.64 $u^2$=0.36 |
| Item 3 Dedication | 0.90 | 0.81 | 0.19 | $h^2$=0.81 $u^2$=0.19 |
| Item 4 Dedication | 0.88 | 0.77 | 0.23 | $h^2$=0.77 $u^2$=0.23 |
| Item 5 Vigor | 0.80 | 0.64 | 0.36 | $h^2$=0.64 $u^2$=0.36 |
| Item 6 Absorption | 0.85 | 0.72 | 0.28 | $h^2$=0.72 $u^2$=0.28 |
| Item 7 Dedication | 0.74 | 0.55 | 0.45 | $h^2$=0.55 $u^2$=0.45 |
| Item 8 Absorption | 0.79 | 0.62 | 0.38 | $h^2$=0.62 $u^2$=0.38 |
| Item 9 Absorption | 0.65 | 0.42 | 0.58 | $h^2$=0.42 $u^2$=0.58 |

Average communality $h^2$ = 0.640 | Total variance explained by 1 factor = 64.0% (averaged across items)
Note: Paper reports > 70% total variance explained by the unrotated 1-factor solution.

# EFA Factor Loading Matrix — 1-Factor Unrotated Solution

All 9 UWES-9 items load strongly (> 0.65) on a single Work Engagement factor




## Loading Matrix Λ (p×m, m=1)

$$\Lambda = \begin{array}{l} \text{Item 1 (V)} \\ \text{Item 2 (V)} \\ \text{Item 3 (D)} \\ \text{Item 4 (D)} \\ \text{Item 5 (V)} \\ \text{Item 6 (A)} \\ \text{Item 7 (D)} \\ \text{Item 8 (A)} \\ \text{Item 9 (A)} \end{array} \begin{bmatrix} \mathbf{0.77} \\ \mathbf{0.80} \\ \mathbf{0.90} \\ \mathbf{0.88} \\ \mathbf{0.80} \\ \mathbf{0.85} \\ \mathbf{0.74} \\ \mathbf{0.79} \\ \mathbf{0.65} \end{bmatrix}$$

## Key Statistics

| | |
|---|---|
| **Range of $\lambda$:** | 0.65 – 0.90 |
| **Min loading:** | 0.65 (Item 9, Absorption) |
| **Max loading:** | 0.90 (Item 3, Dedication) |
| **χ² fit (1-factor):** | 332.43, df=27, p<0.001 |

**Reproduced Correlation (element i,j):**

$$\hat{r}_{ij} = \lambda_i \times \lambda_j$$

Example: Item 1 × Item 3:
0.77 × 0.90 = 0.693 (observed: 0.62)

**All 9 loadings exceed the 0.65 threshold**
**→ EFA strongly supports a SINGLE factor**
**representing overall "Work Engagement"**

# Step 5: Factor Retention Criteria

Multiple tests applied — most converge on 1 factor

| Criterion | Method | Decision | Agrees with 1F? |
|---|---|---|---|
| **Kaiser Criterion ($\lambda > 1$)** | Keep factors with eigenvalue > 1<br>$\lambda_1 \approx 6.38 > 1 \rightarrow \lambda_2 \approx 0.95 < 1$ | **1 Factor** | ✓ |
| **Scree Plot (Cattell, 1966)** | Retain factors before the "elbow" — sharp drop in eigenvalue curve | **1 Factor** | ✓ |
| **Velicer's MAP Test (Original)** | Minimize average squared partial correlation.<br>MAP = $\Sigma r^{2}_{pp} / p(p-1)$ | **1 Factor** | ✓ |
| **Velicer's MAP Test (Revised, 2000)** | Uses 4th-power partial correlations.<br>More accurate in simulation studies. | **1 Factor** | ✓ |
| **Parallel Analysis (Horn, 1965)** | Compare $\lambda_i$ vs. 95th percentile from random data. Retain if $\lambda_i > \lambda_i^{n3rm}$. | **4 Factors** | ✕ |

### Why Parallel Analysis Disagrees (Over-factoring)

Parallel Analysis is known to over-factor when inter-item correlations are very high (0.52–0.85 here). With multicollinear items, even random data eigenvalues are inflated, leading to spuriously high factor counts. 4 of 5 criteria support 1 factor — the MAP test is most reliable here.

# Step 6: Forced 3-Factor Promax Rotation

Testing the theoretical 3-factor structure — items misload onto wrong subscales

## Promax Rotation: Oblique Rotation Allowing Factor Correlations

Start with varimax (orthogonal): $\mathbf{\Lambda}^* = \mathbf{\Lambda T}$ where $\mathbf{TT}^T = \mathbf{I}$ Then raise loadings to power k (k=3 default): $\Lambda^* \rightarrow P$

Result: Pattern matrix P (unique contribution of each factor) + Structure matrix S = P Φ (Φ = interfactor correlation matrix)

### Pattern Matrix: Forced 3-Factor Promax Solution

| Item | Subscale (Theory) | Factor 1 (Dedication+) | Factor 2 (Vigor) | Factor 3 (Absorption) | |
|---|---|---|---|---|---|
| Item 1 | Vigor | 0.03 | **0.79** | 0.01 | |
| Item 2 | Vigor | 0.04 | **0.81** | 0.02 | |
| Item 3 | Dedication | **0.88** | 0.05 | 0.02 | |
| Item 4 | Dedication | **0.86** | 0.03 | 0.03 | |
| Item 5 | Vigor | **0.52** | **0.42** | -0.05 | ⚠ **MISLOAD** |
| Item 6 | Absorption | **0.55** | 0.02 | 0.39 | ⚠ **MISLOAD** |
| Item 7 | Dedication | **0.68** | 0.12 | 0.01 | |
| Item 8 | Absorption | 0.07 | 0.04 | **0.77** | |
| Item 9 | Absorption | -0.02 | 0.06 | **0.72** | |

### Inter-Factor Correlation Matrix Φ

$$\Phi = \begin{vmatrix} 1.00 & 0.80 & 0.83 \\ 0.80 & 1.00 & 0.79 \\ 0.83 & 0.79 & 1.00 \end{vmatrix}$$

$r(F_1, F_2) = 0.80$, $r(F_1, F_3) = 0.83$, $r(F_2, F_3) = 0.79$

All intercorrelations > 0.79 → factors are NOT distinct

### Conclusion from 3-Factor Solution

Items 5 & 6 cross-load across two factors (misloading)

Dedication gains an extra item (Items 5 & 6 load on $F_1$)

Vigor loses one item (Item 5 does not clearly load on $F_2$)

Factor intercorrelations 0.79–0.83 violate distinctiveness

⟹ The 3-factor structure is NOT supported by EFA

# EFA Summary — All Evidence Points to 1 Factor

Five of six tests support a single "Work Engagement" factor (EFA n = 341)

| Test / Analysis | Key Finding | Verdict |
|---|---|---|
| **Correlation Matrix** | r range 0.52–0.85, no clear block structure | ✓ Supports 1F |
| **Eigenvalue Analysis** | $\lambda_1 \approx 6.38$ (>70% variance), $\lambda_2 < 1$ | ✓ Supports 1F |
| **ML Extraction** | $\chi^2 = 332.43$ (df=27), high communalities (0.42–0.81) | ✓ 1-Factor |
| **Scree Plot** | Clear "elbow" after factor 1 | ✓ Supports 1F |
| **Velicer MAP (orig+rev)** | Both versions minimize at m=1 | ✓ 1 Factor |
| **Parallel Analysis** | Raw eigenvalue > 95th percentile for 4 factors | × Suggests 4F |
| **Factor Loadings** | All $\lambda$ = 0.65–0.90, no item clearly multi-factorial | ✓ 1-Factor |
| **Promax 3-Factor** | Items misload; F intercor. = 0.79–0.83 | ✓ Against 3F |

**EFA Conclusion: 7 of 8 analyses support a single "Work Engagement" factor. The Parallel Analysis disagreement is explained by known over-factoring behaviour in datasets with high inter-item correlations (0.52–0.85). The 3-factor Promax solution produces misloaded items and near-collinear factors (r = 0.79–0.83), contradicting the theoretical 3-subscale structure.**

# CFA: Confirmatory Factor Analysis Overview

Three models tested on the holdout sample (n = 342)

## CFA Model Comparison (Holdout Sample n = 342)

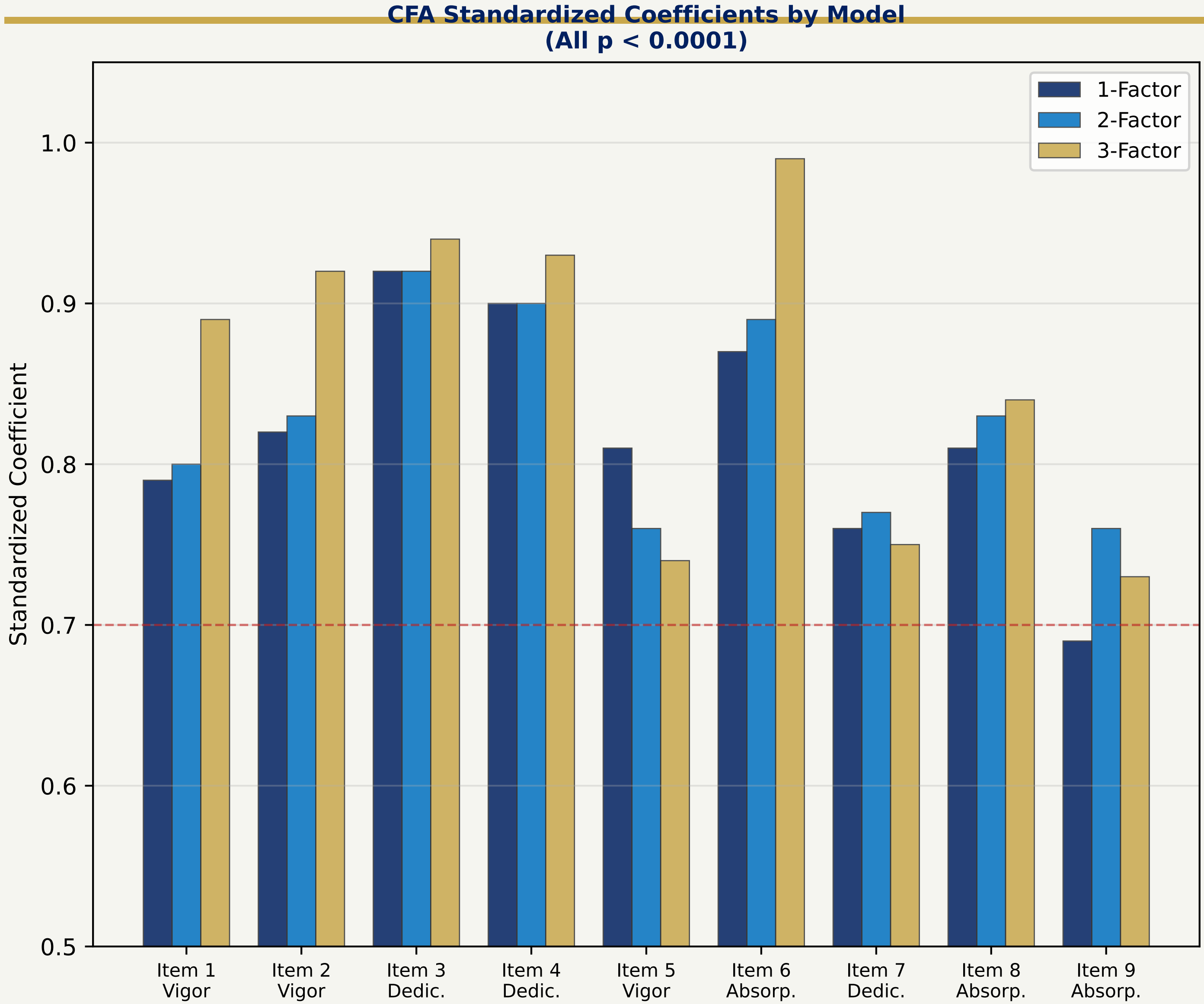
| Statistic | Threshold | Model 1: 1-Factor | Model 2: 2-Factor | Model 3: 3-Factor |
|---|---|---|---|---|
| $\chi^2$ (df) | Non-sig. | **633.90 (27)** | **354.49 (26)** | **247.76 (24)** |
| RMSEA | < 0.05/0.08 | **0.181** | **0.192** | **0.167** |
| CFI | > 0.95 | **0.895** | **0.882** | **0.920** |
| TLI | > 0.95 | **0.860** | **0.837** | **0.880** |
| SRMR | < 0.08 | **0.046 ✓** | **0.049 ✓** | **0.065 ✓** |
| AIC | Lower=better | 16221.47 | 8246.29 | 8143.56 |
| BIC | Lower=better | 16343.70 | 8353.66 | 8258.60 |

**CFA Conclusion: None of the 3 models achieved acceptable fit**

RMSEA ranges 0.167–0.192 (threshold < 0.08) | CFI ranges 0.882–0.920 (threshold > 0.95) | TLI ranges 0.837–0.880 (threshold > 0.95)
Only SRMR met its threshold for all models. CFA cannot confirm the EFA 1-factor solution, nor does it support the theoretical 3-factor structure.

# CFA Standardized Coefficients — All Three Models

All z-values significant ($p < 0.0001$); coefficients stable across models




## Coefficient Table (Table 4)

| Item | 1-F | 2-F | 3-F | z (1F) |
|---|---|---|---|---|
| Item 1 | 0.79 | 0.80 | 0.89 | 50.42 |
| Item 2 | 0.82 | 0.83 | 0.92 | 59.95 |
| Item 3 | 0.92 | 0.92 | 0.94 | 132.99 |
| Item 4 | 0.90 | 0.90 | 0.93 | 109.15 |
| Item 5 | 0.81 | 0.76 | 0.74 | 55.85 |
| Item 6 | 0.87 | 0.89 | 0.99 | 83.55 |
| Item 7 | 0.76 | 0.77 | 0.75 | 44.83 |
| Item 8 | 0.81 | 0.83 | 0.84 | 57.54 |
| Item 9 | 0.69 | 0.76 | 0.73 | 33.19 |

# Discussion — Why Did CFA Fail to Confirm EFA?

Interpreting the EFA vs. CFA discrepancy

## Gender Effects

All-female sample may experience work engagement differently than mixed-gender validation samples. Most prior UWES-9 studies used predominantly male samples (e.g., Hallberg & Schaufeli: 63% male).

## Cultural Context

Swedish work culture emphasizes egalitarianism and work-life balance. Cultural context affects how engagement dimensions are experienced and may flatten distinctions between Vigor, Dedication, and Absorption.

## Instrument Overlap

Inter-factor correlations of 0.79–0.84 suggest the 3 dimensions are nearly identical. Shirom (2003) argued the UWES dimensions were not theoretically deduced and overlap conceptually.

## Sample Sensitivity

Wefald et al. (2012): similar n (≈382), RMSEA = 0.18 and 0.16 — almost identical to our 0.181 and 0.167. The poor fit may reflect instrument limitations, not sample-specific issues.

**The EFA–CFA discrepancy is unlikely to resolve with this dataset alone. Future research should use Bifactor models and Multigroup CFA to test measurement invariance across gender, nationality, and occupation.**

# Conclusion

Summary of EFA & CFA findings | Implications | Future Research

### EFA Findings (n = 341)

- KMO = 0.922 (Marvellous)
- $\lambda_1$ explains > 70% variance
- All loadings $\lambda$ = 0.65–0.90
- MAP test: 1-factor
- Scree: clear single factor
- 3-factor: items misload
- $\Longrightarrow$ 1-factor best fits data

### CFA Results (n = 342)

- Model 1 (1F): RMSEA=0.181
- Model 2 (2F): RMSEA=0.192
- Model 3 (3F): RMSEA=0.167
- CFI: 0.882–0.920 (< 0.95)
- TLI: 0.837–0.880 (< 0.95)
- SRMR: 0.046–0.065 (OK)
- $\Longrightarrow$ No model confirmed

### Key Takeaway & Next Steps

- 1-factor scoring recommended
- for all-female samples
- Gender-specific validation needed
- Bifactor modeling proposed
- Multigroup CFA needed
- Test UWES-3 shorter version
- $\Longrightarrow$ Factor structure unresolved

The optimal factor structure of the UWES-9 in all-female samples remains unresolved. Both EFA and CFA results suggest total (1-factor) scoring is more appropriate than 3-subscale scoring.

# Factor Analysis Methods Applied — Full Summary

All EFA & CFA techniques used in this study

| Method | Purpose | Key Result | Phase |
|---|---|---|---|
| KMO Test | Sampling adequacy | KMO = 0.922 ("Marvellous") | EFA Pre-check |
| Bartlett's Test | $H_0$: R = I (sphericity) | $p < 0.001$ (Reject $H_0$) | EFA Pre-check |
| EFA (ML) | Factor extraction (1-factor) | 1 factor, >70% variance | EFA Core |
| Eigenvalue Decom. | $R = V\Lambda V^T$ spectral decomp. | $\lambda_1 >> \lambda_2$ | EFA Math |
| Scree Plot | Visual λ inspection | Elbow after factor 1 | EFA Retention |
| Velicer MAP (orig) | Minimize avg sq. partial r | 1 factor optimal | EFA Retention |
| Velicer MAP (rev.) | 4th-power partial correlations | 1 factor optimal | EFA Retention |
| Parallel Analysis | Compare λ vs. random data 95th% | 4 factors (over-factors) | EFA Retention |
| Promax Rotation | Oblique 3-factor rotation test | Items misload; r(F)=0.79–0.83 | EFA Rotation |
| CFA: 1-Factor | Test unidimensional model | RMSEA=0.181, CFI=0.895 | CFA |
| CFA: 2-Factor | Test Vigor+Dedic / Absorp split | RMSEA=0.192, CFI=0.882 | CFA |
| CFA: 3-Factor | Test V / D / A structure | RMSEA=0.167, CFI=0.920 | CFA |
| Cronbach's α | Internal consistency | α = 0.947 (Excellent) | Reliability |

# References

All sources cited in this presentation — with full DOI / URL links


### PRIMARY STUDY

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# Thank You

## Questions & Discussion

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SSIE-605: Applied Multivariate Data Analysis

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Primary paper: Willmer et al. (2019) — Frontiers in Psychology
https://doi.org/10.3389/fpsyg.2019.02771